

[ITPP5112]

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Question 1

Q1.1 Four IT Professional Roles Required for the Project

The Move With Me project involved co designing and building a wearable technology prototype a superhero cape with motion sensors, LED lights, Bluetooth Low Energy (BLE) and a solar rechargeable battery together with rural mothers and quilters in Sweet Waters, KZN (Dewlok et al., 2025). Four IT professional roles were needed to make this project succeed.

Project Manager

A)Role in the ICT environment:A project manager plans, organises and controls ICT projects. They manage timelines, budgets, resources, risks and communication between stakeholders, making sure deliverables are completed on time and within scope.

B)Application to the project:This project ran seven co-creation workshops between 31 October 2024 and 16 January 2025, involving two quilters, up to eight mothers and two researchers per session (Dewlok et al., 2025). A project manager would have been responsible for scheduling these workshops, securing the fabrics, hardware and tools, managing the R200 per session honorarium payments, and coordinating the partnership with the governmental research institution and its community liaisons.

C) Community projects involve many non technical stakeholders with their own time pressures (these mothers are informally employed and sacrifice income to attend). Without someone managing logistics, costs and expectations, the trust built with the community can easily break down, and the project would fail regardless of how good the technology is.

Systems/Business Analyst

A)Role in the ICT environment:A systems analyst bridges the gap between users and developers. They gather and analyse user requirements, study business problems, and translate user needs into technical specifications.

B)Application to the project: The researchers used a fully functional prototype cape as a "technology probe" to elicit the mothers' design requirements, and then analysed workshop data (drawings, feedback, prototype interactions) using a reflexive thematic approach (Dewlok et al., 2025). This is systems analysis in action, capturing needs such as offline operation, low cost and the ability to choose cheaper LED lights, and converting them into design changes.

C) In a community driven project, the users' lived experience is the most important requirement source. The analyst role ensures the community's voice is captured accurately and not filtered through the designers' own assumptions which is exactly what prevented a costly, Western style product being imposed on this community.

Software Developer (Mobile Application Developer)

A)Role in the ICT environment:Software developers design, code, test and maintain software applications. They work with programming languages, APIs and platforms to build working products.

B)Application to the project: The team developed the PlayWithMe companion mobile application in Kotlin on the Android Open Source Project platform, chosen because Android holds an 84% market share in South Africa. The app communicates with the cape via a REST API over the BLE protocol and gamifies movement with cartoon characters that react when the child performs the correct movement (Dewlok et al., 2025).

C)Communitydriven projects only create real impact if the final solution actually works for people.

A developer who builds for the devices the community already owns (their Android phones) rather than assuming new hardware keeps the solution affordable and adoptable.

Hardware/Embedded Systems Engineer

A) Role in the ICT environment: An embedded systems engineer designs and builds the electronic components inside devices selecting microcontrollers, sensors, power supplies and communication modules, and programming them to work together.

B) Application to the project: The cape integrates an ADXL345 accelerometer to detect movement on three axes, an ESP32 microcontroller with BLE, a 5V Neopixel LED strip, a DF Robot mini MP3 player with a speaker, and a rechargeable lithium ion battery pack. Movement thresholds had to be calibrated to tell real jumps from false triggers (Dewlok et al., 2025). The mothers were also taught to programme the microcontrollers and adjust voltages during the workshops.

C) Why essential in low resource communities, hardware choices directly determine whether a solution survives in the real world. The engineer's decision to use locally sourceable, low power components meant the cape could be repaired locally and charged with solar power something a purely software focused team would have missed.

Q1.2 Three Essential IT Professional Traits

Three traits stand out as most essential for the team working with the rural mothers and quilters.

Empathy and cultural sensitivity. Working with a low income, Zulu speaking community 97 km from Durban requires professionals who genuinely try to understand the community's world rather than impose their own. The researchers showed this throughout, they paid an honorarium because they understood the mothers' time had income value, they partnered with a local institution that employed community liaisons who knew the community well, and they allowed participants to decide that their images not be blurred because they were "proud of their work" (Dewlok et al., 2025). Without empathy, participants would have been treated as data sources instead of design partners, and the co-design process would have collapsed.

Communication. Technical concepts like microcontrollers, voltages and accelerometers were alien to most mothers at first. The team gave detailed presentations explaining the electronics and their functions before asking mothers to work with them (Dewlok et al., 2025). Good communication turned fear into confidence one mother admitted, "In the beginning I was scared of connecting the wires to the battery and to the charging port. But now I am not scared, it was really not hard." Explaining, reaffirming consent before every photo, and keeping conversations open were all acts of skilled communication.

Patience and adaptability. The researchers openly accepted that the first cape design did not meet the mothers' needs the probe was built precisely to start conversations, not as a finished answer (Dewlok et al., 2025). The team then adapted the design repeatedly based on feedback: more lights, colourful pictures instead of black, fruit and emoji prints on the fabric, and moving sound from the cape to the mothers' phones. Some mothers even disliked working with batteries ("I don't like the batteries, I am scared of them"), and the team accommodated different comfort levels rather than forcing one path. This flexibility is what allowed the mothers to take real ownership of the design.

Q1.3 Strategic Personal Development Plan

A) Two technical skills I must develop, with justification from the paper

Embedded systems and microcontroller programming (C++) The heart of the Move With Me cape is its electronics the ESP32 microcontroller, ADX accelerometer and LED strips, all programmed and calibrated to detect real movement versus false jumps (Dewlok et al., 2025). During the workshops, mothers programmed microcontrollers and discussed the code base. As a junior

graduate, I could not meaningfully contribute to sessions like the 12 December 2024 workshop (programming microcontrollers and adding the accelerometer) without this skill. Developing it would let me help participants troubleshoot and eventually teach others.

Mobile application development with Bluetooth Low Energy (Kotlin/Android):

The PlayWithMe app is the gamification layer that makes children want to move, and it connects to the cape over BLE using a REST API (Dewlok et al., 2025). The mothers gave rich feedback on the app wanting to choose characters, add animal sounds, and even teach ABCs and counting through movement. Implementing any of that requires solid Android and BLE skills, which I would need to develop from a basic graduate level.

B) Two professional development actions aligned with career planning

First, I would enrol in a structured certification path: a Kotlin/Android development course (such as the Google Android Developer certification) combined with an embedded systems course covering Arduino/ESP32, completed within the next 12 months. Second, I would join a professional body such as the ACM and attend their conferences and local chapter events in HCI and ICT for development, which would expose me to co design methodologies and connect me with mentors working in community based research.

C)How these improve employability

These skills and actions improve employability in two ways. Technically, graduates who can work across both hardware and software are far more valuable for development projects, which rarely have the budget for large specialised teams the Move With Me project needed people who could handle sensors, apps and workshops all at once. Professionally, certifications and body membership signal commitment to growth, while conference networking and HCI knowledge (co design, probes, ethics) matches the growing demand in the ICT sector for professionals who can build technology for underserved communities, not just for wealthy urban markets (Dewlok et al., 2025). This combination makes me useful both to NGOs and to corporates with CSI and rural digitalisation agendas.

Question 2 The Professional Body: ACM

The Move With Me paper was published by the Association for Computing Machinery (ACM) in the Proceedings of the 12th International Conference on Communities & Technologies (C&T '25) (Dewlok et al., 2025). ACM is the world's largest scientific and educational computing society, and it functions as the professional body for computing professionals.

Vision, mission and goals. ACM's mission is "to advance computing as a science and profession" (ACM, 2025). Its goals flow from this, promoting technical innovation and high quality research through its journals, conferences and digital library, supporting education through curricula, scholarships and learning resources, fostering professional development and career growth for members, providing a code of ethics that guides professional conduct, and promoting diversity, equity and inclusion so that computing benefits all of society (ACM, 2025). Publishing community based research like the Move With Me paper directly serves the goal of advancing computing as a science it recognises that valuable knowledge is produced outside elite laboratories.

Role in promoting professionalism and ethical standards. ACM sets and maintains a widely respected Code of Ethics and Professional Conduct, which gives ICT practitioners shared principles for responsible behaviour (ACM, 2018). It also influences public policy, accredits educational programmes, and provides forums where researchers and practitioners are held to peer review standards. In this way ACM raises the quality and trustworthiness of the ICT sector as a whole, ensuring that computing work serves the public good rather than only commercial interests.

Two principles from the ACM Code of Ethics applicable to the project.

Principle 1.1 Contribute to society and to human well being, acknowledging that all people are stakeholders in computing. This principle should guide every stage of the project. During co design activities it meant genuinely sharing decision making power with the mothers and quilters rather than token consultation. During data gathering it meant collecting only what was needed, obtaining informed consent (HREC 378/2021), reaffirming it before photos, and honouring participants' wish not to have their images blurred. During prototype testing it meant ensuring the final cape (target price ZAR300-600) actually improved children's physical activity in a low-income community, and donating the capes to the mothers (ACM, 2018; Dewlok et al., 2025).

Principle 1.6 Respect privacy. The project recorded workshops, took photographs and stored transcripts on the institution's cloud behind two-factor authentication. This principle requires that such data be protected diligently, minimised, and used only for the stated research purpose exactly what the team did by anonymising transcripts, restricting access, and allowing participants to withdraw their data at any time (ACM, 2018; Dewlok et al., 2025). Following this principle protects vulnerable participants, especially where children are indirectly involved.

Do bodies like ACM provide value to IT professional researchers?

In my view, yes, they provide real value though not without limitations. On the positive side, bodies like ACM give researchers credibility and quality control. Peer review at conferences such as C&T ensures that findings like Dewlok et al.'s meet scholarly standards before they can influence practice, which protects communities from half baked interventions. ACM also spreads knowledge: a rural co design study published in its proceedings becomes visible to thousands of researchers worldwide, multiplying its impact. Its code of ethics gives researchers a shared moral framework, which matters greatly when working with vulnerable groups such as low income mothers and young children. Membership further provides networking, career resources and a sense of professional identity.

However, there are criticisms worth acknowledging. Membership fees, conference costs and paywalled publications can exclude researchers from poorer institutions and developing countries ironically, the very people doing community based ICT research like this study. Some argue that bodies move too slowly and are dominated by large tech interests.

Even with these flaws, the value outweighs the drawbacks. Without ACM, the Move With Me research would likely have stayed a local report; through ACM it became part of the global conversation on designing for low resource contexts. For IT professional researchers, such bodies remain essential for quality, ethics and reach.

Question 3 IT, Society and Transformation Essay

Technology does not succeed or fail on its features alone it succeeds when it fits the society it serves. The Move With Me project, which co designed a sensor filled superhero cape to encourage physical activity among rural South African children, is a strong example of how ICT can be shaped around community realities rather than imposed on them (Dewlok et al., 2025). This essay discusses the positive impacts of the cape, the skills barrier in rural communities, transformation and inclusion in the ICT sector, the value of co designing with mothers and quilters, and practical initiatives to close the digital divide.

The cape can have at least two positive impacts on rural communities. First, it addresses childhood physical inactivity, which is rising even in areas where children lack safe play spaces and well resourced Early Childhood Development centres. By turning jumping and movement into a game with lights, sounds and cartoon characters, the cape makes exercise fun and can support healthy physical development from an early age. Second, it carries economic potential. The components are cheap and locally sourceable, the mothers have already discussed forming a cooperative with sewing, electronics and C++ programming teams, and marketing the capes at a local fair (Dewlok et al., 2025). What started as a research prototype could become a community run small business.

However, a lack of technical skills limits how fully rural communities can benefit. Although the mothers completed a six month co-design readiness phase with a 45-sensor kit, and grew confident enough to sew in LEDs and connect batteries, several still feared components like lithium batteries, and none could yet programme the ESP32 without support (Dewlok et al., 2025). Without ongoing skills transfer, the community remains dependent on outside researchers for repairs, code changes and app updates. The same gap appears in data and device management: mothers spend R50-R350 a month on mobile data and rely on pre paid electricity, so any solution that assumes technical confidence, unlimited data or stable power will exclude them, no matter how well designed it is.

On transformation and inclusion, the project shows what meaningful transformation looks like. Rural communities are usually the last to receive technology and the first to be priced out of it. Here, the design deliberately targeted ZAR300-600 per cape, works fully offline via BLE, runs on phones the mothers already own, and can be charged with solar power (Dewlok et al., 2025). Just as importantly, the project transferred real capability: mothers moved from being "users" to being makers who can teach others.

Their plan for a cooperative including a team that wants to learn C++ is genuine inclusion in the ICT sector, not charity.

Involving rural mothers and quilters was central to making the cape suitable and culturally grounded, rather than a Western product with local branding. The mothers rejected expensive Neopixel strips in favour of cheaper, bendable lights that let them sew flowers and hearts; they moved sound from the cape to their phones, cutting cost and battery drain; they chose Batman and Superman fabrics their children love, and quilters chose traditional fabrics for grandchildren (Dewlok et al., 2025). One mother suggested the cape teach "ABCs or counting when the child jumps," blending movement with education in a way no outside designer would have imagined. This is the core lesson, co design produced a cheaper, more sustainable, more loved product.

To reduce the digital divide between rural and urban communities, IT professionals could implement two practical initiatives. The first is a community tech hub with affordable hardware and connectivity for example, solar powered Wi-Fi hotspots at the local research institution or schools, together with a device lending library and low cost repair services using the same locally available components as the cape. This directly addresses the cost and connectivity barriers the mothers face. The second is structured upskilling, running free, ongoing workshops where mothers and youth learn electronics, coding and app maintenance, possibly certified through TVET colleges or professional bodies like ACM. In the Move With Me case, the six-month readiness phase transformed participants' confidence; scaling such programmes turns one-off projects into lasting digital capability.

In conclusion, Move With Me demonstrates that ICT transformation in rural communities depends less on advanced technology than on affordability, offline functionality, skills transfer and genuine co design. When communities help build the technology, they do not just receive a product they gain the ability to sustain, repair and even sell it.

Reference List

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